CONTROL SYSTEMS
ENGINEERING PROJECT

Frequency Control of a Single-Area Power System

Using a PI Controller and Solar PV Generation in MATLAB/Simulink

PROJECT SNAPSHOT

Nominal frequency	Controller	Simulation
50 Hz	PI control	MATLAB/Simulink 2026
Disturbances	Stability method	Simulation horizon
Load & solar PV	Routh-Hurwitz	100 seconds

PROJECT TEAM

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A simulation study of how a conventional generation system responds to load and photovoltaic power disturbances, and how PI control helps restore nominal frequency.

01 | Introduction

The objective of this work is to model and simulate a single-area power system comprising a conventional generator and solar photovoltaic (PV) generation. The study investigates the system's frequency response to load and PV disturbances under proportional–integral (PI) control.

The nominal frequency is 50 Hz. The model was built and simulated in MATLAB/Simulink 2026 as a linear, small-signal model using per-unit (pu) deviation variables. The controller was tuned analytically and validated through time-domain simulation.

The control target

Maintain $\Delta f \approx 0$ despite step changes in load (ΔP_L) and PV output (ΔP_{PV}). In practical terms, the frequency should return to 50 Hz with a damped transient and zero steady-state error.

02 | Why frequency changes

Frequency deviation reflects the instantaneous mismatch between electrical generation and demand. When demand exceeds generation, the system experiences a power deficit and frequency falls. When generation exceeds demand, frequency rises.

PV generation is inverter-interfaced and does not provide rotational inertia in the same way as a conventional synchronous generator. Its output can also change with irradiance and cloud cover, so PV variation acts as an additional disturbance. A higher PV share can lower the effective inertia of the system.

Power balance used in the model

$$\Delta P_{net} = \Delta P_m + \Delta P_{PV} - \Delta P_L$$

$$\Delta f(s) = G_{gen}(s) \cdot \Delta P_{net}(s)$$

03 | System model and parameters

The conventional generation chain and generator–load dynamics are represented by first-order transfer functions.

Transfer functions

Governor: $G_{gov}(s) = 1 / (0.2s + 1)$

Turbine: $G_{tur}(s) = 1 / (0.5s + 1)$

Generator–load: $G_{gen}(s) = 1 / (2Hs + D)$

With H = 5 s and D = 1 pu: $G_{gen}(s) = 1 / (10s + 1)$

Droop gain: $1/R = 1/0.05 = 20$

Model parameters

Parameter	Symbol	Value
Nominal frequency	f_0	50 Hz
Inertia constant	H	5 s
Load damping	D	1 pu
Droop	R	0.05 pu (gain 20)
Governor time constant	Tg	0.2 s
Turbine time constant	Tt	0.5 s
Generator–load time constant	2H/D	10 s
Simulation time	Tsim	100 s

04 | PI controller design

A parallel-form continuous PI controller was used. Its gains were obtained with MATLAB’s pidtune(Gref, 'PI') function and then checked through simulation.

$C(s) = Kp + Ki/s$

$C(s) = 1 + 9.89387/s = (s + 9.89387)/s$

Controller parameter	Value
Proportional gain (Kp)	1
Integral gain (Ki)	9.89387

A preliminary reference-to-frequency step-response comparison reported a lower overshoot for the PI controller (0.55%) than for the integral controller (1.32%). Both responses converge to the reference.

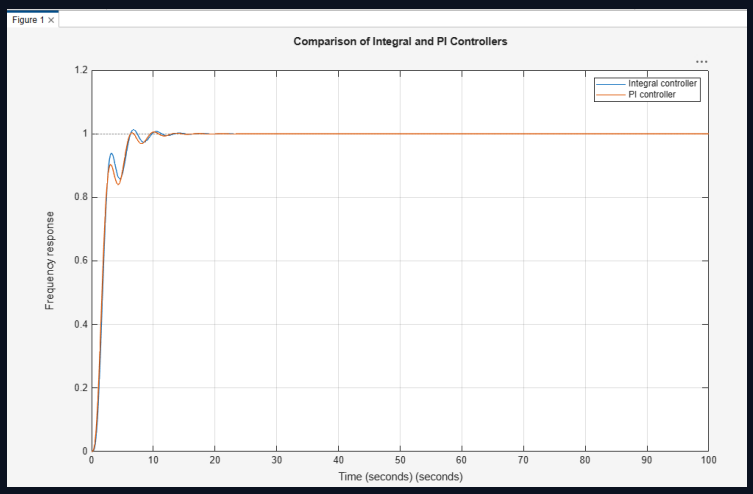


Figure 1. Step response: integral controller vs. PI controller.

05 | Closed-loop analysis

The reference is set to zero and the net disturbance is $d = \Delta P_{PV} - \Delta P_L$. The governor input combines PI action and droop feedback.

$$u = -C(s)\cdot\Delta f - 20\cdot\Delta f$$

$$\Delta P_m = G_{gov}\cdot G_{tur}\cdot u = -G_{gov}\cdot G_{tur}\cdot(C(s) + 20)\cdot\Delta f$$

$$\Delta f = G_{gen}\cdot(\Delta P_m + d)$$

$$\Delta f/d = G_{gen} / [1 + G_{gen}\cdot G_{gov}\cdot G_{tur}\cdot(C(s) + 20)]$$

After numerical substitution, the disturbance-to-frequency transfer function is:

$$T(s) = \Delta f/d = (0.1s^3 + 0.7s^2 + s) / (s^4 + 7.1s^3 + 10.7s^2 + 22s + 9.89387)$$

06 | Stability: Routh-Hurwitz criterion

The Routh array for the characteristic polynomial is shown below. All entries in the first column are positive, so there is no sign change. Based on this test, the closed-loop system is stable.

Row	Column 1	Column 2	Column 3
s^4	1	10.7	9.89387
s^3	7.1	22	0
s^2	7.601	9.89387	0
s^1	12.76	0	0
s^0	9.89387	0	0

First-column values: 1, 7.1, 7.601, 12.76, 9.89387.

07 | Poles and damping

The roots of the characteristic polynomial are $s_1 = -5.866$, $s_2 = -0.546$, and $s_{3,4} = -0.344 \pm j1.723$. The dominant complex pair is lightly damped, which explains the oscillatory transient.

$$\text{Natural frequency: } \omega n = \sqrt{(0.344^2 + 1.723^2)} = 1.757 \text{ rad/s}$$

$$\text{Damping ratio: } \zeta = 0.344 / 1.757 = 0.196$$

$$\text{Damped period: } Td = 2\pi / 1.723 = 3.6 \text{ s}$$

08 | Steady-state error

For a step disturbance d_0/s, the final value theorem gives:

$$\Delta f(\infty) = \lim_{s\to 0} s \cdot T(s) \cdot (d_0/s) = T(0) \cdot d_0$$

With PI control, T(0) = 0 / 9.89387 = 0, so Δf(∞) = 0.
For droop control alone (without integral action), the report gives:

$$\Delta f(\infty) = d_0 / (D + 1/R) = 0.01 / 21 = 4.76 \times 10^{-4} \text{ pu}$$

$$\text{In Hz: } 4.76 \times 10^{-4} \times 50 = 0.0238 \text{ Hz}$$

09 | MATLAB/Simulink implementation

The signal path in the Simulink model is:

Reference (0) → error sum (reference − Δf) → PI → droop sum → governor → turbine → power-balance sum (+ + −) → generator–load → output Δf.

The droop feedback (gain 20) is taken from Δf and subtracted at the droop sum. The PV step source is connected to the positive input and the load step source to the negative input of the power-balance sum. Δf is displayed on a Scope and saved by a To Workspace block as the timeseries df_load.

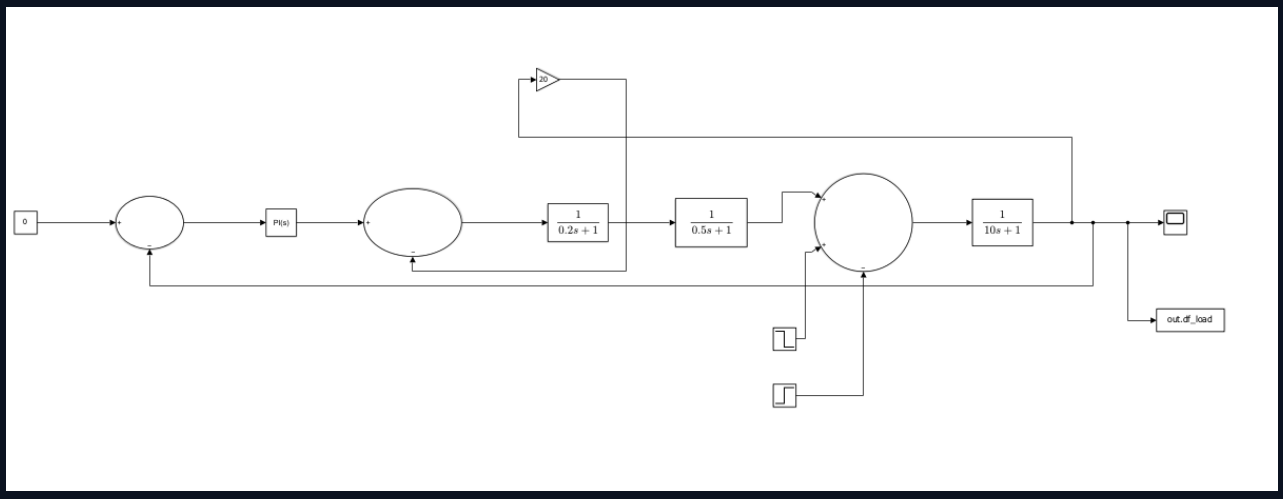


Figure 2. Simulink block diagram of the single-area model.

10 | Simulation scenarios

All scenarios use the same plant and controller (Kp = 1, Ki = 9.89387) over 100 seconds. Disturbances are expressed in pu; frequency deviation in Hz is calculated as Δf(Hz) = 50 × Δf(pu).

Experiment	Step time	Disturbance	Net power d
1 – Load increase	1 s	ΔPL: 0 → +0.01 pu	−0.01 pu
2 – Load decrease	1 s	ΔPL: 0 → −0.01 pu	+0.01 pu
3 – PV increase	1 s	ΔPPV: 0 → +0.01 pu	+0.01 pu
4 – PV fluctuation	1 s & 20 s	+0.01 then −0.01 pu	+0.01 then −0.01 pu

11 | Experiment results

Experiment 1 – Load increase. A 0.01 pu deficit causes an initial frequency drop. The peak is −0.0338 Hz at 2.06 s, followed by an opposite peak of +0.0082 Hz at 4.02 s. The response enters the 2% settling band at 11.70 s, which is 10.7 s after the disturbance.



Figure 3. Load increase (+0.01 pu at t = 1 s).

Experiment 2 – Load decrease. A 0.01 pu surplus produces the mirror response: the peak is +0.0338 Hz at 2.06 s, followed by −0.0082 Hz at 4.02 s. The 2% settling time is 10.7 s and the steady-state deviation is zero.



Figure 4. Load decrease (−0.01 pu at t = 1 s).

12 | PV disturbance results

Experiment 3 – PV increase. A +0.01 pu PV step is equivalent to the load-decrease case in the power balance. The peak frequency deviation is +0.0338 Hz at 2.06 s, followed by an undershoot of −0.0082 Hz at 4.02 s. The response settles in approximately 10.7 s.

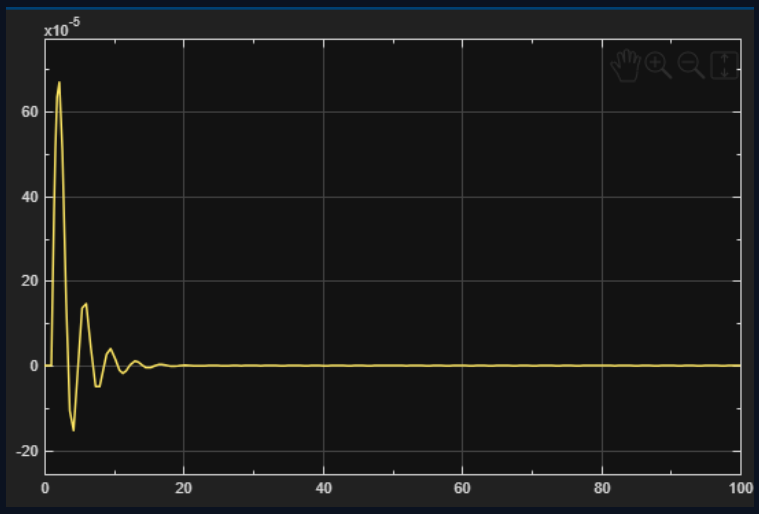


Figure 5. PV increase (+0.01 pu at t = 1 s).

Experiment 4 – PV fluctuation. PV generation first increases by 0.01 pu at 1 s and then decreases by 0.01 pu at 20 s. The first response has settled by the time the second step occurs. The second disturbance produces a peak of approximately −0.0338 Hz at 21.06 s, followed by a damped response.



Figure 6. PV fluctuation (+0.01 pu at 1 s, −0.01 pu at 20 s).

Summary of reported metrics

Experiment	Peak Δf (pu)	Peak Δf (Hz)	Opposite peak (pu)	Settling time
Load +0.01	-6.76×10^{-4}	−0.0338	$+1.64\times10^{-4}$	10.7 s
Load −0.01	$+6.76\times10^{-4}$	+0.0338	-1.64×10^{-4}	10.7 s
PV +0.01	$+6.76\times10^{-4}$	+0.0338	-1.64×10^{-4}	10.7 s
PV +0.01 / −0.01	$+6.76$ / -6.75×10^{-4}	+0.0338 / −0.0338	$-/+1.64\times10^{-4}$	10.7 s each

Note from the source report: the summary metrics are marked “Values to be confirmed against df_load.” Experiment 5 (combined disturbance) was excluded because signal routing had not been verified.

13 | Discussion

A load increase creates a net power deficit ($d < 0$) and produces an initial frequency drop. A PV increase creates a net surplus ($d > 0$) and produces an initial frequency rise. A load decrease and a PV increase of equal magnitude are equivalent in the power balance and therefore produce identical responses in this linear model.

The PI controller eliminates steady-state error in all tested cases. By comparison, droop-only control leaves a steady-state deviation of 4.76×10^{-4} pu for the 0.01 pu step disturbance. The damping is the same across the tested scenarios because the model is linear and the disturbance magnitude is constant.

The oscillation comes from the interaction between governor–turbine response lags and system inertia. Droop limits the initial frequency deviation, while integral action restores the nominal 50 Hz frequency.

14 | Conclusion

The model demonstrates primary frequency control using a PI controller and droop feedback under load and solar PV disturbances. The closed-loop system is stable according to the Routh-Hurwitz criterion. In all tested scenarios, the frequency returns to the nominal 50 Hz with damped transients, a reported peak deviation of 0.0338 Hz, a settling time of approximately 10.7 seconds, and zero steady-state error.

The results apply to the small-signal disturbances and model assumptions used in this study.

15 | Reference

Das, D.C., Sinha, N., Roy, A.K. (2014). “Automatic Generation Control of an Organic Rankine Cycle Solar–Thermal/Wind–Diesel Hybrid Energy System.” Energy Technology, 2, 721–731. DOI: 10.1002/ente.201402024.

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